

Mathematical Foundations ANSWER KEY

November 30, 2017

Instructions

This exam has 6 sections. Provide the answers to each section on a new page, labeling the part of the question for each answer.

I PROBABILITY AND MATHEMATICAL LOGIC

- 1) If $P(A \cap B) = 0$, which of these are false? (there may be more than one)
 - a) $P(A|B) = P(A)$ **FALSE** : If $P(A \cap B) = 0$, A and B are *mutually exclusive*. This would only be true if A and B were *independent*.
 - b) $P(A|B^c) = \frac{P(A)}{P(B^c)}$ **TRUE** : $P(A|B^c) = \frac{P(A \cap B^c)}{P(B^c)}$ and $P(A) = P(A \cap B^c) + P(A \cap B)$.
 - c) $P(A) + P(B) = P(A \cup B)$ **TRUE**: $P(A) + P(B) - P(A \cap B) = P(A \cup B)$
- 2) Why is it that:
 - a) If $A \subset B$ then $P(B|A) = 1$ **Answer**: $A \cap B = A$, so $P(B|A) = \frac{P(A \cup B)}{P(A)} = \frac{P(A)}{P(A)}$
 - b) If $A \subset B$ then $P(A|B) = \frac{P(A)}{P(B)}$ **Answer**: $A \cap B = A$, so $P(A|B) = \frac{P(A \cup B)}{P(B)} = \frac{P(A)}{P(B)}$
- 3) Roll three six-sided dice once.
 - a) What is the probability of no six? **Answer**: $(5/6)^3 = .57$
At least one six? **Answer**: $1 - (5/6)^3 = .42$
At least two sixes? **Answer**: Either there are two six or three six: $\frac{1}{6} * \frac{1}{6} * \frac{5}{6} + \frac{1}{6} * \frac{5}{6} * \frac{1}{6} + \frac{5}{6} * \frac{1}{6} * \frac{1}{6} + \frac{1}{6}^3 = 0.074$
 - b) Find the probability that there are at least two sixes given that there is at least one six. (hint: use the equation in question 2.b). **Answer**: $0.074/0.42 = 0.176$
- 4) TRUE or FALSE, if false, explain in one sentence.
 - a) $\bar{X} = \mu$ **Answer: FALSE** The sample mean is not equal to the population mean. $\bar{X}$ is a random variable.
 - b) If X and Y are independent, $Var(X - Y) = Var(X) + Var(Y)$ **Answer**: This is true (and in the book).
 - c) $\bar{X} \sim N(0, 1)$ **Answer: FALSE** Only if you divide by the standard deviation and subtract the mean.
 - d) $Var(\mu) = 0$ **Answer: TRUE**
 - e) $Var(\bar{X}) = \frac{\sigma^2}{n}$ **Answer: TRUE**

II BASKETBALL!

Since 2010, the Women's National Basketball Association has a best of five series for the National Championship. This means that the series can end only when one team wins 3 games. We are interested in the number of games that are played for one team to be named Champion.

- 1) If both teams are equally likely to win, what is the probability that the series lasts to exactly 4 games? **$3*2*(1-p)*p*p*p = 3/8$**
- 2) What is the probability that the series lasts less than 5 games? **This is one minus the chance that it lasts 5 games: $1-3/8 = 0.625$**
- 3) What is the expected number of games played? **$1/4*3+3/8*4+3/8*5=4.125$**

III TRIANGLE DISTRIBUTION

Suppose that X is a random variable distributed as follows:

$$f(x) = \begin{cases} 0 & \text{for } x < 0 \\ 4x & \text{for } 0 \leq x < 1/2 \\ 4 - 4x & \text{for } 1/2 \leq x \leq 1 \\ 0 & \text{for } 1 < x \end{cases}$$

- 1) Draw $f(x)$. **This looks like a triangle.**
- 2) Verify that $f(x)$ is a pdf. **It needs to integrate to one.**
- 3) Draw the CDF of X . **This is a line**
- 4) What is the probability that $X > .7$? **Answer: $\int_{.7}^1 4 - 4x dx = 4x - 2x^2|_{.7}^1 = 2 - 1.82 = .18$**

IV SURVEYING SENSITIVE INFORMATION.

It is difficult to conduct surveys on sensitive issues because many people will refuse to give answers that might embarrass them or put them in legal trouble. "Randomized response" is a method designed to ensure anonymity when asking about such topics. Suppose you wish to determine the prevalence of illegal drug use in a population. Each respondent is asked to privately flip a coin. If they get a tails, they must say 'yes', whether or not they had used drugs. If they get heads, they should tell the truth and say 'no' if they had not used illegal drugs. Only the respondent knows whether the answer reflects the truth or the coin toss.

- 1) List the possible outcomes. {not do drugs, say no} etc.
- 2) Suppose 25% of the respondent population uses illegal drugs. What is the probability of a 'no'? **Answer: A no arises if the coin lands heads and they didn't use drugs. so $.5*.75 = .375$**
- 3) Now suppose that you conduct the study and find that 35% of the respondents give a 'no' answer. What would be your estimate of the percent of respondents who have used illegal drugs? **Answer: 30% used illegal drugs.**

V CALCULATING EXPECTATIONS

- 1) Suppose X and Y are random variables that are jointly distributed

$$f(x, y) = \begin{cases} \frac{16y}{x^3}, & \text{if } x > 2, 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

- a) What is the formula for $f(x)$? What is $f(y)$? **Answer:**

$$f(x) = \int_0^1 \frac{16y}{x^3} dy = \frac{8}{x^3}$$

$$f(y) = \int_2^\infty \frac{16y}{x^3} dx = -(-(\frac{16y}{2^3})) = 2y$$

- b) What is $E[X]$? **Answer:** $\int_2^\infty x \frac{8}{x^3} = \int_2^\infty \frac{8}{x^2} = \frac{8}{x} \Big|_2^\infty = 0 - \frac{-8}{2} = 4$

- c) What is $E[X|Y = .5]$? How about $E[X|Y = .21]$? $E[X|Y = .98]$? **Answer:** X and Y are *independent* because $f(x, y) = f(x) * f(y)$ so they all =4.

- 2) Suppose that the sample space S contains three elements $\{1, 2, 3\}$, with probabilities .5, .2, and .3 respectively. Define a random variable X as follows:

$$X(s) = s^2 - 4 \text{ for } s \in S$$

- a) What is $E[X]$? **Answer:** $.5*(1-4) + .2*(4-4) + .3*(5) = 0$

- b) What is $\text{Var}[X]$? **Answer:** $.5 * (-3 - 0)^2 + .2 * (0 - 0)^2 + .3 * (5 - 0)^2 = 12$

- 3) Suppose you enter a casino, and they offer you one of two bets $\{\mathbf{A}, \mathbf{B}\}$

A Roll a six-sided die 5 times, with sides printed $\{1, 2, 3, 4, 5, 6\}$, take the average of your rolls and receive the number of dollars equal to the average, translated by the following equation:

$$g(\bar{x}) = e^{\bar{x}-3.5}$$

B Roll a six-sided die 5 times, where each of the sides of the die has $e^{x-3.5}$, $x \in \{1, 2, 3, 4, 5, 6\}$ written on it. i.e. $\{e^{-2.5}, e^{-1.5}, e^{-.5}, e^{.5}, e^{1.5}, e^{2.5}\}$, and receive the average of the number of dollars on its face.

- a) Without calculating the exact value, which of these bets would you take, **A** or **B**? Why? **I prefer B because of Jensen's inequality. $g(x)$ is convex.**

- b) What if $g(\bar{x})$ were $\sqrt{\bar{x}}$? What if $g(\bar{x})$ were $\log(\bar{x})$? Again you shouldn't need to calculate the expectation to know the answer. **In these cases, I prefer A, as $g(x)$ would be concave. You can know this by taking the second derivative or by recognizing the shape of these famous functions.**

VI LIMIT THEOREMS

- 1) Suppose $\bar{X}$ is the sample mean of 100 observations from a population with mean μ and variance σ^2 . Find 90% confidence limits for $\bar{X}$.

a) Using Chebychev.

$$P(|\bar{X} - \mu| \geq c) \leq \frac{\sigma_{\bar{x}}^2}{c^2}$$

Choosing $c = \sigma_{\bar{x}}k$, we get the following:

$$P(|\bar{X} - \mu| \geq k\sigma_{\bar{x}}) \leq \frac{1}{k^2}$$

so $.1 = \frac{1}{k^2}$, $k = 3.16$

$$P(|\bar{X} - \mu| \geq 3.16\sigma/\sqrt{100}) \leq .1$$

$$P(\bar{X} - 3.16\sigma/\sqrt{100} \geq \mu \geq \bar{X} + 3.16\sigma/\sqrt{100}) \leq .1$$

The confidence interval is $\pm .316\sigma$

b) Using Central Limit Theorem.

$$P\left(\frac{|\bar{X} - E(X)|}{\sigma/\sqrt{100}} \geq \Phi^{-1}(.95)\right) = .1$$

$$P(|\bar{X} - E(X)| \geq 1.64\sigma/\sqrt{100}) = .1$$

The confidence interval is $\pm 0.164\sigma$.

2) Suppose $\bar{X}_1$ and $\bar{X}_2$ are two independent surveys of size n from the same population. Assume we further know that for the individual responses, $Var(X_i) = \sigma^2$.

a) What is the distribution of $\bar{X}_1 - \bar{X}_2$? (hint: see problem 4 from section I).

Answer: $N(0, \sigma^2/n + \sigma^2/n) = N(0, 2\sigma^2/n)$

b) Suppose you had to pick a sample size such that:

$$P(|\bar{X}_1 - \bar{X}_2| < \sigma/5) \approx .99$$

Use the fact that $\Phi^{-1}(.995) = 2.576$ to decide how big of a sample you should use.

Given the distribution of $\bar{X}_1 - \bar{X}_2$ is normal, we have:

$$P\left(\frac{|\bar{X}_1 - \bar{X}_2 - 0|}{\sqrt{2}\sigma/\sqrt{n}} \leq \Phi^{-1}(.995)\right) = .99$$

$$P(|\bar{X}_1 - \bar{X}_2| \leq \sqrt{2}\sigma * \Phi^{-1}(.995)/\sqrt{n}) = .99$$

$$\sqrt{2}\sigma * \Phi^{-1}(.995)/\sqrt{n} = \sigma/5$$

$$\sqrt{2}\Phi^{-1}(.995)/\sqrt{n} = 1/5$$

$$\sqrt{2} * 2.576/\sqrt{n} = 1/5$$

$$\sqrt{2} * 2.576 * 5 = \sqrt{n}$$

$$n = (\sqrt{2} * 2.576 * 5)^2 = 332$$